

Warm-up 1

a) Write down the matrix that maps the standard basis $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ to the vectors $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 2 \end{pmatrix}$, respectively.

b) What vectors does this matrix send to $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$?

During class 1

a) Using the standard basis, the vector v is given as $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$. Write v using the basis $b_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $b_2 = \begin{pmatrix} -2 \\ -2 \end{pmatrix}$, where these new basis vectors are also given in the standard basis.

b) Using the standard basis, a linear transformation is given by the matrix A . Write the same linear transformation using v_1 and v_2 as an ordered basis for $\mathbb{R}^2$.

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad v_2 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$